**XSSV Challenge II: Phase 2 - Feasibility Design & Pilot Development**
**Congratulations on Your Selection!**
Your team's proposal has been selected as one of the winning submissions for XSSV Challenge II! The XSSV partners were impressed by your innovative approach to utilizing excess solar energy and carbon dioxide to create valuable chemical products.
You have been awarded **$500,000** to develop a comprehensive feasibility design and prepare for pilot-scale demonstration of your concept. This funding will support your work over the next 12 months as you transition from concept to a detailed technical and economic analysis.

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**Email from XSSV Partners**
**From:** XSSV Partners (partners@xssv.com)
**Subject:** XSSV Challenge II - Phase 2 Award and Requirements

Dear Challenge II Winners,
Congratulations on your selection! Your team stood out among dozens of submissions for your innovative thinking and well-reasoned approach to addressing our excess solar and $CO_2$ utilization challenge.
As we move into Phase 2, we need you to validate your concept with rigorous technical and economic analysis. The $500,000 award is intended to support:

- Detailed process design and engineering analysis
- Market research and business model development
- Economic analysis including capital and operating costs
- Environmental impact assessment
- Risk analysis and mitigation strategies

At the end of this phase, we will evaluate all winning teams and select 2-3 projects to advance to Phase 3 (Pilot Development) with additional funding of $5-10 million each.

**Your Phase 2 Deliverables:**
We expect a comprehensive feasibility design report proposal that addresses the technical, economic, and strategic viability of your proposed process. This should build upon the preliminary work you completed in your Challenge II submission. You will present your proposed idea in an open poster session on December 3 to interested parties.
We look forward to seeing how your concept develops into a viable pathway for commercial deployment.

Best regards,
XSSV General Partners

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# Assignment: Feasibility Design Report
Working with your team, develop a comprehensive feasibility design for your XSSV Challenge II winning concept. Your analysis should demonstrate whether your process can be technically successful, economically viable, and environmentally beneficial.
**Written Report Requirements (5-7 pages)**
Your feasibility design report should include the following sections:
**1. Executive Summary**

- Brief overview of your project
- Key findings and recommendations
- Required next steps for commercialization

**2. Technical Design**

a. **Process Description & Chemistry**
   - Chemical reactions
   - Detailed Block Flow Diagram with major unit operations
   - Operating conditions of unit operations (temperature, pressure, ect)

- Mass and/or Energy balances for key streams with flow of materials in and out of each unit

**b. Material Selection**

- Raw materials: sources, purity requirements, cost, specifically how much CO2 will your process use
- Product specifications and purity targets
- Byproducts and waste streams
- Storage and handling considerations

**c. Scale & Capacity**

- Proposed production capacity (kg/hr or tonnes/year)
- Justification based on market demand and available excess solar

**3. Economic Analysis**
**Revenue Projections**

- Product selling price (with market research justification)
- Annual production volume
- Revenue from any byproducts
- **Total Annual Revenue (Product sales – raw material costs)**

**4. Market & Business Strategy Market**
**Analysis**

- Updated market size and growth projections
- Competitive landscape analysis (who else is making this product)

**5. Environmental & Sustainability Assessment**
**Life Cycle Considerations**

- Carbon footprint analysis ($CO_2$ utilized vs. emissions from process)
- Net greenhouse gas reduction compared to conventional production
- Water usage and wastewater generation
- Circular economy opportunities

**6. Safety**
**Risk Analysis**

- What safety issues may occur in your process?
    - Use information from SDS, such as toxicity, flammability, ect… anywhere in your process (raw materials, products, by products, waste, ect)
- What other risks can happen in your process and How you plan to address each major risk

**7. Recommendations & Next Steps**

- Is the project technically feasible?
- Is the project economically viable?
- Should XSSV invest in Phase 3 (Pilot Development)?
- Critical experiments or analyses needed before pilot-scale

**8. References & Appendices**

- Properly cited sources (technical literature, market reports, cost data) using IEEE referencing format
- Supporting calculations
- Additional BFD/PFDs, equipment specifications, or technical details

## Report Evaluation Criteria

Your feasibility design will be evaluated on:

**Technical Rigor (30%)**

- Soundness of chemistry and engineering principles
- Completeness of process design
- Appropriate use of calculations and simulations
- Realistic assessment of technical challenges

**Economic Analysis (25%)**

- Cost estimations
- Revenue projections
- Appropriate use of financial metrics

**Market Understanding (15%)**

- Depth of market research
- Realistic competitive assessment
- Clear commercialization strategy

**Environmental Impact (15%)**

- Quantitative sustainability analysis
- Comparison to alternatives
- Understanding of full life cycle

**Critical Thinking & Communication (15%)**

- Quality of risk analysis
- Clarity of recommendations
- Professional presentation
- Ability to defend assumptions under questioning

# Poster Presentation

Using the poster template, prepare a poster for the XSSV Investment Committee in an open-forum style with all other winners present to provide questions and feedback. Your presentation should be tailored for an audience that includes:

- Chemical engineers
- Business development professionals
- Venture capital partners
- Sustainability and policy advisors

**Suggested Poster Topics:**

1. Recap of your concept
2. Technical design highlights
    - Key reactions and process flow
    - Major innovations or technical advantages
3. Economic viability
    - Capital and operating costs
    - Revenue model and profitability
    - Sensitivity analysis
4. Market opportunity
    - Market size and competitive positioning
5. Environmental impact
    - $CO_2$ reduction potential
    - Sustainability metrics
6. Recommendations
    - Go/No-Go decision with justification
    - Required next steps

**Presentation Tips:**

- Use clear, professional visuals
- Include process diagrams and economic charts
- Be prepared to defend your assumptions
- Address potential investor concerns proactively

- Demonstrate that you understand both the opportunities and challenges

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**Looking Ahead: Phase 3**

Teams that demonstrate strong technical feasibility and economic viability will be invited to Phase 3, where you will:

- Build and operate a pilot-scale system
- Generate data for commercial-scale design
- Engage with potential customers and partners
- Prepare for Series A venture funding

This is your opportunity to transform an innovative idea into a real solution for climate change and sustainable chemical production. Good luck!